

Sangam G Sajjan

QA Automation Engineer

Bangalore, Karnataka • +91 7483138916 • sangamgsajjan102@gmail.com

Summary

QA Automation Engineer with experience in Playwright, Python, and API automation. Skilled in building scalable test frameworks, AI-assisted test generation, CI/CD integration. Passionate about delivering reliable, maintainable automation that accelerates releases and improves software quality.

Project

AI-Augmented E2E Automation Ecosystem [Git | Demo](#)

- Architected an end-to-end automation framework using **Playwright Python and Pytest** for CRUD validation.
- Engineered a reusable **Page Object Model framework** with **custom fixtures** and centralized locator management,
- **custom fixtures** reducing maintenance effort by approximately **60%** across **regression** suites.
- Leveraged **GitHub Copilot prompt** engineering to auto-generate edge-case assertions and increase script velocity.
- Integrated Playwright automation into **GitHub Actions** with **parallel execution** and automated **HTML reports** for every **pull request**.
- Containerized Playwright automation using **Docker** to ensure consistent execution across **developer** and **CI** environments.

AI-Assisted API & Synthetic Data Testing Pipeline [Git | Demo](#)

- Engineered an **automated API testing** pipeline using **Playwright APIRequestContext** to validate **RESTful endpoints**.
- Integrated **OpenAI API** workflows to parse schemas and **auto-generate negative** and **boundary-value** payloads.
- Implemented **JSONSchema** validation in Python to enforce strict structural verification on response bodies.
- Utilized **Pytest fixtures** to mock external dependencies, ensuring isolated and reliable test suite execution.

Experience

ENGINEER QA

SEP 2024 – PRESENT

IDS NEXT BUSINESS SOLUTIONS PVT LTD

REMOTE

- Owned manual testing across **13 integrated modules** for a cloud application, covering release readiness and defect prevention.
- Extracted **600+ functional test cases** from **BRDs** using **prompt engineering** to improve requirements coverage and reduce test design effort.
- Automated requirements-to-test-case generation using **LLM**, reducing manual authoring effort by **30%** while improving edge-case coverage.
- Investigated development defects using **JIRA** and **Excel**, **SQL** validation, and **root cause analysis**,
- accelerating issue resolution and preventing recurring **API data integrity** issues.
- Coordinated QA activities across major product releases, aligning testing with cross-functional teams to ensure on-time deployments

Skills

- **Programming & Databases:** Python, SQL, MySQL
- **Test Automation:** Playwright, Pytest, Page Object Model (POM),
- **API Testing:** REST API Testing, Postman, Playwright API Testing, JSON Schema Validation
- **DevOps, Cloud & Version Control:** JIRA, Browser Developer Tools, Git, GitHub Actions, Docker, AWS
- **AI:** GitHub Copilot, OpenAI API, Prompt Engineering,
- **Manual Testing:** STLC, SDLC, Functional Testing, Smoke Testing, Sanity Testing, Regression Testing, UI Testing, Cross-Browser Testing, Responsive Testing, Compatibility Testing, Exploratory Testing, Boundary Value Analysis

Education

Kalpataru Institute of Technology | VTU University

May 2024

BE in Computer Science -- 76%